

1 Abstract

2 1. The collation of thousands of population projection matrices in the COM(P)ADRE
3 Matrix Databases has enabled large-scale comparative analyses in ecology, evolution, and
4 demography. A persistent challenge is that transition rates are estimated from finite
5 samples, yet the resulting sampling uncertainty is rarely reported and typically ignored in
6 downstream analyses. Although sampling uncertainty is well recognised, it remains
7 unclear when it affects demographic inference.

8 2. Using a subset of studies from the COMPADRE Plant Matrix Database for which
9 transition-specific sample sizes could be obtained, we reconstructed sampling
10 distributions for transition rates and derived demographic parameters, and conducted

11 three related analyses: (i) a multi-species comparative analysis of life expectancy-shape
12 relationships, (ii) a single-species time-series analysis of climate-demography effects, and
13 (iii) a simulation study with known truth that compares how models account for count
14 sampling and year-to-year variation.

15 3. The effects of sampling uncertainty were strongly trait-dependent. Accounting for
16 sampling uncertainty barely shifted integrated metrics such as population growth rate,
17 but substantially displaced traits related to mortality trajectories, particularly the shape of
18 the mortality distribution. In these cases, point estimates were often systematically
19 displaced from their underlying sampling distributions due to boundary estimates of
20 stage-specific survival (i.e. survival probabilities near 0 or 1), which are common in
21 published matrices. By contrast, inference in a within-species analysis of weather effects
22 on vital rates was largely unaffected, whereas the simulation study showed that point-
23 estimate count models can become overconfident even when bias remains low.

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4. Together, these analyses show that ignoring sampling uncertainty does not universally compromise inference from published projection matrices, but can yield fragile or biased conclusions for certain demographic traits; the simulation also shows that omitting year-to-year variation can produce overconfident interval estimates in count-based models.

We identify structural features of MPMs and analyses that may indicate heightened sensitivity to sampling uncertainty, and discuss implications for researchers using compiled demographic databases.

Keywords: age-from-stage methods; estimation bias; parameter uncertainty; stage-structured models; sensitivity analysis; plant demography; life history traits

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46 Introduction

47 Sampling uncertainty is a widely recognised, inherent component of demographic analyses
48 (Caswell 2001). This uncertainty, stemming from the sampling of the population of interest,
49 proliferates across almost all estimates of individual demographic rates (e.g. survival and
50 reproduction), through studies based on raw data and single species, up to large comparative
51 studies.

52 The collation of published demographic data in databases such as PanTHERIA (Jones et
53 al. 2009), AnAge (de Magalhães and Costa 2009), Myhrvold et al.'s (2015) amniote life-history
54 database, and COM(P)ADRE (Salguero-Gómez et al. 2015, Salguero-Gómez et al. 2016a) has
55 enabled fascinating comparative analyses in ecology, evolution, and demography (e.g. Garratt et
56 al. 2013, Adler et al. 2014, Healy et al. 2014, Fristoe et al. 2017). By leveraging open-source
57 databases that aggregate data from hundreds of individual studies, researchers can test
58 hypotheses of interest across taxonomic, spatial, and temporal scales that would otherwise be
59 impossible. However, analyses based on published demographic results rather than underlying
60 raw data also present their own challenges.

61 One challenge in working with large collections of published data is understanding and
62 modelling the sources of underlying uncertainty. For instance, the demographic parameters
63 reported in the databases noted above have almost always been estimated from a sample of
64 individuals drawn from a larger population of interest, and therefore carry inherent sampling
65 uncertainty. However, parameter-specific sample sizes and standard errors are rarely available
66 and are often difficult to obtain even from the original data sources. These original studies,
67 which produce the parameter estimates later compiled in demographic databases, usually account
68 for sampling uncertainty in their own analyses, but uncertainty is often not reported in a form
69 that can be propagated from the published matrices. Therefore, studies that subsequently analyse
70 parameter estimates compiled in demographic databases rarely incorporate this uncertainty,

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71 raising questions about the robustness of resulting inferences (Caswell 2001, Ch. 12; Biggs et
72 al. 2009).

73 Despite widespread recognition that sampling uncertainty is inherent in demographic estimates
74 (Caswell 2001), a key unresolved question concerns when this uncertainty materially affects
75 inference from compiled databases. Comparative analyses using projection matrices rely on a

76 diverse set of derived demographic traits; population- or species-level quantities computed from

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77 matrix population models (MPMs). These span life-history traits such as generation time,

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78 transient descriptors such as reactivity, and integrated metrics such as population growth rate,

79 through to traits governed by long-term survivorship or tail behaviour, such as life expectancy

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80 and the shape of mortality trajectories. Whether these different classes of traits are equally

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81 sensitive to sampling uncertainty under typical data regimes remains poorly understood.

82 Recent work has reviewed the importance of uncertainty propagation in matrix population
83 models and outlined best-practice guidelines for addressing it (Simmonds and Jones 2024). That
84 paper focused on methodology and characterised the sources of uncertainty, the available
85 propagation techniques, and recommendations for practice, but did not empirically quantify the
86 consequences of ignoring uncertainty across a range of real analyses and demographic traits.
87 Here, we address that complementary question directly: not how uncertainty should be handled,
88 but when handling it actually changes the conclusions researchers draw from published
89 projection matrices.

90 Large comparative analyses are not the only analysis that can be impacted by omission of
91 sampling uncertainty. Another key area where sampling uncertainty could have a substantial
92 impact is in linking demographic rates to abiotic drivers. When incorporating drivers, such as
93 climate, in MPM studies researchers typically regress an annual demographic rate
94 (e.g. recruitment) against a climate variable (e.g. spring temperature) to estimate how strongly
95 climate drives dynamics. A key challenge with this approach is that the vital rate estimates in the

100 regression are themselves noisy. In linear regression, independent, additive, mean-zero error in
101 the response reduces precision but does not itself bias the slope towards zero. Its effects in
102 models of demographic rates depend on the error structure and model formulation; whether
103 explicitly accounting for sampling uncertainty materially changes climate-response inference is
104 therefore an empirical question, which we address here.

105 We investigate sampling uncertainty in the COMPADRE Plant Matrix Database (Salguero-
106 Gómez et al. 2015; www.compadre-db.org), which currently contains over 9,000 matrix
107 population models (i.e. projection matrices, hereafter MPMs) drawn from more than 650 studies.
108 Over the course of its development, researchers have used COMPADRE (and the sister animal
109 matrix database COMADRE; Salguero-Gómez et al. 2016a) to address diverse topics in ecology,
110 evolution, and demography, including studies of the structuring of life-history variation (Franco
111 and Silvertown 1996, Salguero-Gómez et al. 2016b), comparative analyses of ageing (Baudisch et
112 al. 2013, Jones et al. 2014), tests of the demographic buffering hypothesis (McDonald et
113 al. 2017), and analyses of the impact of climate on population dynamics (Coutts et al. 2016,
114 Csergő et al. 2017). In practice, these analyses involve deriving parameters from a set of MPMs
115 in COMPADRE (e.g. equilibrium population growth rate λ , elasticities of λ , reproductive values,
116 mean vital rates, generation time, life expectancy), and ~~then using~~ these parameter estimates,
117 ~~which themselves carry sampling uncertainty inherited from the underlying transition rates, or~~
118 ~~further derivations of them, as inputs to a statistical model that tests the relationship of interest.~~
119 ~~Uncertainty must therefore propagate along a chain, from the sampled transition rates, through~~
120 ~~the derived parameters, to the final inference.~~

121 Our aim is to examine how sampling uncertainty in published projection matrices propagates to
122 specific MPM-derived quantities and to explore when point estimates are robust or fragile. We
123 expect sensitivity to vary systematically across traits, reflecting a spectrum from highly integrated
124 metrics — such as population growth rate, which averages over many transitions and should be

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127 buffered against uncertainty in any single element — to traits that depend on long-term
128 survivorship inferred indirectly via age-from-stage methods, such as life expectancy and mortality
129 shape, where small errors in late-stage survival can amplify into large errors in derived quantities.

130 ~~We combine a comparative analysis of lifetime-integrating traits across populations and species~~
131 ~~(Analysis 1), a within-species analysis of climate effects on demographic rates (Analysis 2), and a~~
132 ~~simulation assessing estimation bias and interval coverage against known climate effects~~
133 ~~(Analysis 3). A multisite extension of Analysis 2 using *Astragalus scaphoides* provides a~~
134 ~~supplementary robustness check.~~ The focal quantities are mature life expectancy, the shape of
135 the mortality trajectory, log population growth rate, and recruitment-climate regression
136 coefficients. Because COMPADRE does not routinely include transition-level sample sizes, our
137 ~~empirical analyses are~~ restricted to the subset for which these data could be assembled (695
138 MPMs from 31 studies; details in Methods). We use these ~~analyses~~ to identify diagnostic features
139 that help researchers assess robustness in their own work. ~~We also provide a practical diagnostic~~
140 ~~checklist for COMPADRE users.~~

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141 **Methods**

142 *Background on the COMPADRE Plant Matrix Database*

143 The COMPADRE database includes MPMs and associated metadata from published studies
144 dating back to 1966 (Salguero-Gómez et al. 2015). Where possible, MPMs are decomposed into
145 component processes of survival, growth, and retrogression (the U matrix), sexual reproduction
146 (F), and clonal reproduction (C). The database provides a range of metadata for each MPM,
147 including the relevant time period, geographic coordinates of the study site, population or site
148 identifiers, and treatment conditions. From a given source study, COMPADRE often contains
149 multiple MPMs reflecting different sites, time periods, and/or treatment conditions, as well as

159 one or more “mean MPM” calculated as an element-by-element arithmetic mean across multiple
160 years and/or sites.

161 *Target studies*

162 We used COMPADRE to identify candidate studies for data collection, with the following
163 criteria: published from 2010 onwards; common perennial growth forms (herbaceous, shrub,
164 tree, palm, and succulent); wild populations not subjected to experimental treatments; and
165 MPMs with three or more stage classes and a one-year projection interval. This yielded a dataset
166 of 31 studies, representing 695 MPMs for 132 populations of 30 species, including a small
167 number added under separate criteria (Supplementary Methods section S1.8). Studies were most
168 often excluded because transition-specific sample sizes were insufficiently reported, with a few
169 using estimation methods incompatible with our uncertainty-reconstruction approach (e.g. mark-
170 recapture or multi-stage vital-rate products). We compared the included studies with the other
171 studies that met the same criteria but lacked usable sample sizes, and found them broadly similar
172 in matrix dimension, life form, and mean stage-specific survival, although our included studies
173 had a higher proportion of boundary survival estimates, especially survival of exactly 1 (Table
174 S11). This over-representation may favour detection of boundary-related sensitivity and limits
175 generalisation to all MPM studies.

176 *Typical construction of matrix population models*

177 A matrix population model A is a square projection matrix whose elements A_{ij} represent
178 estimated transition rates from stage class j to stage class i over one projection interval (typically
179 one year). Transitions may reflect the passage of individuals from one stage to another through
180 survival, growth, or retrogression (the U component), or the production of new individuals via
181 reproduction (the F component); $A = U + F$. Sometimes the reproductive transitions are
182 further split into sexual and clonal contributions.

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- Deleted: temporal replication), giving a final set of 31 studies representing 695 MPMs for 132 populations of 30 species

209 The methodology for estimating transition rates varies across studies. In plant MPM studies,
 210 survival-related transitions are often estimated from observed stage transitions of marked
 211 individuals between censuses, as in Eq. (1); in other systems, especially mobile animals, transition
 212 and survival estimation more commonly uses models that explicitly account for imperfect
 213 detection.

$$214 \quad U_{ij,t} = \frac{\text{number transitioned from stage } j \text{ in year } t-1 \text{ to stage } i \text{ in year } t}{\text{number in stage } j \text{ in year } t-1} \quad (1)$$

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215 Reproductive transitions are sometimes estimated directly from counts of new recruits at time t
 216 per reproductive individual at time $t-1$, as in Eq. (2),

$$217 \quad F_{sr,t} = \frac{\text{number in recruit stage } s \text{ in year } t}{\text{number in reproductive stage } r \text{ in year } t-1} \quad (2)$$

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218 The demographic interpretation of F depends on census timing and stage definitions (e.g. pre-
 219 breeding vs post-breeding formulations). Here, Eq. (2) is a count-based estimator of the recruit
 220 transition across one projection interval and we use the published matrix entries as represented
 221 in COMPADRE rather than reparameterising each study to a single census convention.

222 Alternatively, reproductive transitions may be decomposed into lower-level vital rates
 223 (e.g. flowers per plant, seeds per fruit, and germination probability) estimated from separate
 224 samples, or transition rates may be fitted with models that partially pool estimates across stage
 225 classes, sites, or years, as in hierarchical vital-rate models and integral projection models (IPMs).
 226 Uncertainty in such models can differ from the simple count-based setting considered here:
 227 partial pooling often reduces the variance of weakly informed estimates but can induce stronger
 228 correlations across transitions, so uncertainty should ideally be propagated from the joint
 229 posterior of the fitted model. Our reconstructions therefore approximate the uncertainty
 230 available to a secondary user of the published matrix and its sample-size information, rather than
 231 the full joint uncertainty from the original fitted model.

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246 *Sampling distribution for a single MPM*

247 Techniques for estimating sampling uncertainty in transition rates and derived parameters are
248 reviewed in Caswell (2001, Ch. 12). We reconstruct uncertainty using direct draws from
249 conjugate posterior distributions: 2,000 draws per MPM for the illustrative example and 1,000
250 for Analysis 1. Hereafter, “sampling distributions” refers to these reconstructed posterior
251 uncertainty distributions. Survival-related transitions are drawn jointly within each origin-stage
252 column using a Dirichlet prior with all concentration parameters equal to 1, including death;
253 reproductive rates use independent flat priors on the non-negative rate scale (Supplementary
254 Methods section S1.1). For example, if U_{ij} is a survival-related transition from stage j to i ,
255 estimated from the observation that k out of the n individuals in stage j survived and
256 transitioned to stage i (and the remaining $n - k$ died), we can model the U_{ij} transition as a
257 binomial sample with k successes in n trials. We could equivalently use a multinomial likelihood
258 here (given observed counts x_m of each possible outcome m), which flexibly scales to instances
259 with more than two possible outcomes (e.g. an individual from stage j might die, survive and
260 remain in stage j , or survive and grow to stage $j + 1$). For reproductive transitions, we use a
261 Poisson likelihood to model the number of recruits k produced by a given number of
262 reproductive individuals n . Use of a Poisson distribution for fecundity might under-capture
263 overdispersion in species with highly variable fecundity.
264 Because it draws from the posterior rather than resampling the reported estimate, this approach
265 assigns uncertainty to allowed outcomes that were not observed, including death when all
266 sampled individuals survived, so it does not force every draw to reproduce boundary point
267 estimates. The resulting uncertainty depends on the priors and on the allowed-transition
268 structure, which we defined per population; it may be understated for transitions that are
269 biologically possible but never observed for a population, and columns with missing or zero

Deleted: The method we use here is parametric bootstrapping, in which we generate a sampling distribution for each MPM by taking 2,000 independent random draws from the sampling distribution of each underlying transition rate.

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recorded sample sizes contribute no reconstructed uncertainty (Supplementary Methods section S1.1).

Analyses

Illustrative example

We illustrate with an MPM for the perennial forb *Agrimonia eupatoria* (Rosaceae) from a study carried out in southern Sweden (Kiviniemi 2002) (Population ‘B’; projection interval 1997-1998). The full matrix is

$$A = \begin{bmatrix} 0 & 0 & 0 & 0.19 \\ 0.86 & 0.78 & 0 & 0 \\ 0.06 & 0.67 & 0.28 & 0 \\ 0 & 0 & 0.22 & 0.70 \end{bmatrix}.$$

Decomposed into survival/growth (U) and reproduction (F) components:

$$U = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0.86 & 0.78 & 0 & 0 \\ 0.06 & 0.67 & 0.28 & 0 \\ 0 & 0 & 0.22 & 0.70 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & 0 & 0 & 0.19 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

In that study, ~80 plants were monitored each year in a population estimated at ~1000 individuals. Mirroring the $A = U + F$ decomposition above, the observed transition frequencies can be written as:

$$N_U = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 12/14 & 14/18 & 0 & 0 \\ 0 & 1/18 & 6/9 & 13/47 \\ 0 & 0 & 2/9 & 33/47 \\ 2/14 & 3/18 & 1/9 & 1/47 \end{bmatrix}$$

$$N_F = \begin{bmatrix} 0 & 0 & 0 & 9/47 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Deleted: We chose parametric bootstrapping because the sampling distributions of survival and reproductive transitions follow well-characterised probability models (multinomial and Poisson, respectively) whose parameters can be directly estimated from the transition frequencies reported in source studies. This makes parametric bootstrapping more tractable than MCMC for our scale of analysis and more principled than the empirical bootstrap, which cannot generate variation when all individuals in a stage class experience the same outcome (e.g. all survive), a situation common in published matrices. When a stage-specific sample size was recovered as zero, we retained the published point-estimate transition column unchanged in all bootstrap draws rather than treating it as a structural zero. In such cases, the matrix remains usable in the same way it would be in a comparative database analysis, but no additional sampling variation is propagated for that unsupported column.

312 Here, columns index origin stage j ; rows 1-4 in N_U index destination stages, and the extra row d

313 is death. In the posterior sampling, death probability is drawn jointly with the living-transition
 314 probabilities in each U -column, but only the living-stage components are written back into the
 315 simulated U matrix. The N_F matrix remains 4×4 and contains only reproductive transitions.

316 The count likelihoods used to reconstruct transition-rate uncertainty are:

317 $(12,2) \sim \text{Multinomial}(U_{21}, U_{d1}),$

318 $(14,1,3) \sim \text{Multinomial}(U_{22}, U_{32}, U_{d2}),$

319 $(6,2,1) \sim \text{Multinomial}(U_{33}, U_{43}, U_{d3}),$

320 $(13,33,1) \sim \text{Multinomial}(U_{34}, U_{44}, U_{d4}),$

321 $9 \sim \text{Poisson}(47F_{14}).$

322 To create a sampling distribution for the focal MPM, we take 2,000 independent draws from the

323 posterior distributions obtained using these likelihoods and the priors described above (see
 324 Supplementary Methods section S1.1 for a full workflow description). At each iteration, we

325 arrange the drawn transition rates into the form of our MPM and then use that MPM draw to
 326 calculate our derived parameter(s) of interest. After 2,000 iterations, we therefore have 2,000
 327 independent draws from the sampling distribution of each transition rate and derived parameter.

328 The modest sample sizes underpinning this MPM (9-47 plants per stage class) result in relatively
 329 high sampling uncertainty in the transition rates (Fig. 1). For instance, the sample estimate for
 330 the vegetative-reproductive growth transition is 0.22 (2/9), but the upper bound of the 95%

331 credible interval suggests this transition rate could plausibly be as high as 0.45 in the population
 332 at large. Correspondingly, there is substantial sampling uncertainty in some derived parameters

333 (Fig. 2). Based on the 95% credible interval bounds, the equilibrium population growth rate

334 could be as low as -17% per year or as high as +1% per year; life expectancy could be as low as

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352 3.5 years or as high as 18.2 years; and the reproductive value of the vegetative stage class could
353 be 22% lower than that of the seedling stage or 1460% higher.

354 We chose three analyses to ~~examine~~ contrasting ~~settings where~~ sampling uncertainty may ~~matter~~.
355 Analysis 1 is a comparative analysis across many species and populations, in which derived traits
356 depend heavily on long-term survivorship inferred indirectly from stage-structured matrices,
357 precisely the setting in which sampling uncertainty might compound. Analysis 2, by contrast, is a
358 within-species time-series analysis in which the response variable is a directly observed count, the
359 statistical model is closely tied to the raw data, and the question concerns an environmental

360 driver rather than a demographic trait ~~itself~~. Analysis 3 ~~is a simulation study with a known truth,~~
361 ~~designed to test whether these models perform well when year-to-year process variation is~~
362 ~~present across a range of effect sizes and sample sizes.~~ Together, the three analyses ~~cover~~ a
363 ~~broad~~ range of typical uses of published MPMs and ~~examine how modelled sources of~~
364 ~~uncertainty affect inference in different~~ contexts.

365 *Analysis 1: Comparative analysis of mortality trajectories*

366 The analysis we focus on here tests the relationship between mature life expectancy and the
367 “shape” of the mortality distribution (sensu Baudisch 2011), where shape denotes the extent to
368 which mortality rates increase with age after reproductive maturity. Evolutionary biologists have
369 long been interested in variation in mortality trajectories within evolutionary theories of
370 senescence, but previous metrics used to compare rates of senescence may have conflated
371 variation in the shape of mortality trajectories with variation in the average level of mortality
372 (i.e. the inverse of life expectancy). Measures of shape proposed by Baudisch and colleagues
373 (Baudisch 2011, Baudisch et al. 2013, Wrycza et al. 2015) are mathematically independent of life
374 expectancy and therefore well-suited for comparative analyses. Whether shape is biologically
375 independent of life expectancy is a separate question, and an analysis by Baudisch et al. (2013)
376 (based on MPMs for 290 angiosperm species) found a positive relationship between life

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expectancy and the shape of mortality trajectories. In addition to that analysis, several other studies have used estimates of life expectancy and shape derived from MPMs in COM(P)ADRE, primarily in studies of life-history variation (e.g. Salguero-Gómez et al. 2016b; Healy et al. 2019).

To estimate mature life expectancy (L) and shape (S), we first used age-from-stage methods (Cochran and Ellner 1992, Caswell 2001 Sec. 5.3) to derive a survivorship trajectory from each MPM, starting from the first stage class(es) with positive reproduction (see Supplementary Methods section S1.2 for details). We then estimated mature life expectancy by summing over the survivorship trajectory, as in Eq. (3),

$$L = \sum_x l_x, \quad (3)$$

where l_x is the proportion of a cohort surviving from reproductive maturity to age x . To estimate shape, we used a metric proposed by Baudisch and Stott (2019), which compares the area under a survivorship trajectory to that under a reference trajectory assuming constant mortality with age. The metric can take any continuous value between -0.5 (maximal negative senescence; all individuals die at the age of maturity) and $+0.5$ (maximal senescence; all individuals die at the age of maximum longevity), with a value of 0 indicating a constant mortality rate (negligible senescence). Values of shape greater than, equal to, or less than zero correspond to Type I, II, and III survivorship curves, respectively. Thus, more positive values indicate that mortality is concentrated later in adult life, whereas more negative values indicate that mortality is concentrated earlier in adult life. When deriving a survivorship trajectory from an MPM, the apparent mortality rate necessarily flattens as the stage distribution approaches equilibrium (Horvitz and Tuljapurkar 2008). To avoid bias from artefactual mortality plateaus, we calculated shape from truncated survivorship trajectories, limited to the period before a cohort reaches its quasi-stable distribution (based on a distance threshold of 1%; see Supplementary Methods sections S1.2 and S1.13).

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418 We calculated life expectancy and shape at the population level, using one population-level MPM
 419 per population. For populations with temporal replication, we used a pooled population MPM if
 420 available, (i.e. a matrix constructed by pooling individual-level demographic information across
 421 periods and/or populations); otherwise, we used a population mean MPM, if available (i.e. an
 422 element-by-element arithmetic mean across annual matrices); if neither was available, we
 423 constructed a population-level matrix by taking the element-wise mean of the available annual
 424 Individual matrices in COMPADRE. When database records included both site-specific
 425 populations and aggregate population labels (e.g. combined multi-site entries), we retained the
 426 site-specific populations and excluded the aggregate rows to avoid double counting. To model
 427 sampling uncertainty for pooled or mean population-level MPMs, we pooled stage-specific
 428 sample sizes across years (see Supplementary Methods section S1.3). This approach ignores
 429 uncertainty arising from temporal variation in transition rates and therefore underestimates
 430 overall uncertainty in the pooled estimates. The degree of underestimation depends on how
 431 much transition rates vary across years relative to within-year sampling error; for species with
 432 high interannual variability, the bias could be substantial. Results should therefore be interpreted
 433 conservatively with respect to the influence of sampling uncertainty.

434 Our uncertainty calculations used the per-stage or per-transition sample sizes we could recover.
 435 The total estimated population size was not consistently reported across the included studies, so
 436 we did not use it. For 5 of the population matrices (one population each), we could not recover
 437 even the per-stage sample sizes, so we had no basis for estimating their uncertainty. Where these
 438 matrices were kept, the affected columns were held at their published values rather than given
 439 reconstructed sampling uncertainty.

440 To model the relationship between point estimates of mature life expectancy and shape, we used
 441 a linear model with random intercepts at the species level (to account for non-independence
 442 among populations of the same species), as in Eq. (4),

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$$y_{ij} \sim \text{Normal}(\alpha_j + \beta x_{ij}, \sigma^2), \quad (4)$$

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where x_{ij} and y_{ij} represent point estimates of life expectancy and shape, respectively, for population i and species j , α_j represent species-specific intercepts, β is the slope of the relationship between S and L , and σ^2 is residual variance. Species-level intercept variation is modelled as $\alpha_j \sim \text{Normal}(\mu_\alpha, \tau^2)$, where τ^2 is the among-species intercept variance (i.e., variance of random intercepts; distinct from residual variance σ^2). To incorporate sampling uncertainty, we used a measurement error model (based on McElreath 2020, sec. 14.1.2) where latent x_{ij} and y_{ij} are linked to sampling-distribution means (x_{ij} and y_{ij}) and variances ($\sigma_{x,ij}^2$ and $\sigma_{y,ij}^2$) derived as described in the sampling-distribution section above, as in Eqs. (5), (6), and (7).

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The structural form of the model is unchanged when sampling uncertainty is incorporated; Eqs. (5)-(7) restate the outcome model and add explicit sub-models for the true latent values of x_{ij} and y_{ij} , which are now treated as unknown quantities estimated from their **reconstructed posterior** distributions rather than as fixed **point estimates**:

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$$y_{ij} \sim \text{Normal}(\alpha_j + \beta x_{ij}, \sigma^2), \quad (5)$$

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$$x_{ij} \sim \text{Normal}(x_{ij}, \sigma_{x,ij}^2), \quad (6)$$

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$$y_{ij} \sim \text{Normal}(y_{ij}, \sigma_{y,ij}^2). \quad (7)$$

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For these models, we used priors and settings as described in Supplementary Methods section S1.4.

In supplementary analyses, for each derived parameter we compared how much its point estimates varied across population-level MPM entries with how much its posterior means varied. A ratio greater than 1 means the point estimates are more spread out than the posterior means. It does not tell us how much of that spread comes from sampling noise and how much from

Deleted: In supplementary analyses, we estimated the variance components of life expectancy, shape, and other life history traits using a sampling-to-other variance ratio: the proportion of total observed variation in a demographic parameter attributable to transition-rate sampling uncertainty relative to variation from all other sources (principally biological variation among populations and species). For each parameter we partitioned total variance into a sampling component, estimated as the mean of the within-MPM sampling variances derived from the bootstrap distributions, and a residual component representing the remainder. A ratio greater than 1 indicates that sampling uncertainty exceeds biological variation; a ratio less than 1 indicates that biological variation dominates. Full details are given in Supplementary Methods section S1.5 and Supplementary Table S3.4

490 genuine biological differences among populations. We obtained 95% intervals by paired
491 bootstrap resampling of entries (Supplementary Methods section S1.5; Table 1).

492 *Analysis 2: The impact of weather on demographic traits*

493 We focus on a 12-year MPM time-series for *Silene spaldingii* (Lesica and Crone 2007), a perennial
494 herb found in semi-arid grasslands of western North America. We selected this case because it
495 provides a rare combination of multi-year temporal replication, recoverable transition-level

496 sample sizes, and matched local climate covariates (PRISM data extracted at the geographic
497 coordinates of the study population and matched in time to the demographic observations),
498 making it well suited for testing uncertainty propagation in a climate-response context. In that

499 study, transition rates were estimated using the methods described above, as ratios of two
500 integers (Eqs. (1) and (2)), and the corresponding stage-specific sample sizes are reported in Ellis
501 et al. (2012). Eight of the MPMs in the 12-year time-series were non-ergodic due to missing

502 estimates for the “rosette” stage class (i.e. no rosettes were observed in the sample plots). Instead
503 of examining variation in population growth rates (which assumes ergodicity), we focus on the
504 reproductive transition (hereafter ‘recruitment’, the number of seedlings in year t per

505 reproductive plant in year $t - 1$), and survival. We note that recruitment estimated in this way is
506 a relatively aggregated transition whose behaviour under sampling uncertainty may differ
507 substantially from that of survival transitions, which are estimated from smaller stage-specific

508 samples and are more prone to boundary estimates. Any robustness in this analysis would
509 therefore reflect properties of aggregate reproductive transitions under moderate sample sizes,
510 and should not be taken to imply that within-species analyses are always insensitive to sampling
511 uncertainty.

512 We examined the effect of monthly temperature deviations, derived from PRISM monthly
513 temperature rasters extracted at the study population coordinates (Daly et al. 2008; PRISM
514 Group, Oregon State University 2026; Supplementary Methods section S1.6), on annual

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521 recruitment using a similar approach to Teller et al. (2016), in which we model separate slope
 522 coefficients for a series of K monthly lags (going back in time from the end of each transition
 523 interval), smoothing coefficients across neighbouring lags. The smoothing model is a Gaussian
 524 process with an exponential covariance function, as in Eq. (8),

$$525 \quad \log(y_t) \sim \text{Normal}\left(\alpha + \sum_k b_k x_{tk}, \sigma^2\right), \quad (8)$$

526 with the monthly-coefficient prior and covariance structure given in Eqs. (9) and (10),

$$527 \quad b \sim \text{MVN}(\mu_b, K), \quad (9)$$

528 and

$$529 \quad K_{i,j} = \eta^2 \exp(-0.5 \rho^2 (j - i)^2), \quad (10)$$

530 respectively, where y_t is annual recruitment in year t , and x_{tk} is the temperature deviation for
 531 year t at monthly lag k . Model parameters include the intercept (α), residual variance (σ^2),
 532 vector of monthly temperature coefficients (b), mean of the distribution of b (μ_b), covariance
 533 matrix (K), maximum covariance for the b distribution (η^2), and the rate of decay in covariance
 534 with time (ρ^2).

535 In the model above, y_t represents a point estimate of annual recruitment derived from the
 536 relevant MPM. Recall that these values were initially calculated as the number of seedlings in year
 537 t divided by the number of reproductive adults in year $t - 1$. To propagate sampling
 538 uncertainty, we model the raw seedling counts directly with a Poisson likelihood (Eq. (11)) and
 539 include $\log(N_{t-1})$ as an offset via the year-specific intercept model in Eq. (12), so expected
 540 counts scale with the number of reproductive plants sampled. Note that the two formulations
 541 also differ in model structure: the point-estimate model uses a Gaussian likelihood on log per-
 542 capita recruitment, whereas the uncertainty-aware model uses a Poisson likelihood on raw counts

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556 with a log-offset. Observed differences between formulations therefore reflect both structural
 557 change and uncertainty propagation, though in practice the two are virtually identical.

558 Accordingly, we model seedling counts as

559
$$Y_t \sim \text{Poisson} \left(\exp \left(\alpha_t + \sum_k b_k x_{tk} \right) \right), \quad (11)$$

560 and year-specific intercepts as

561
$$\alpha_t \sim \text{Normal}(\mu_\alpha + \log(N_{t-1}), \sigma_\alpha^2). \quad (12)$$

562 This implies that uncertainty in per-capita recruitment declines with sample size: if $r_t = Y_t/N_{t-1}$,
 563 then $\text{Var}(r_t)$ scales as $1/N_{t-1}$ for fixed underlying recruitment intensity.

564 We used the same coefficient prior and covariance structure as for the point-estimate model

565 (Eqs. (9) and (10)), where the response variable Y_t is the number of seedlings in year t , and N_{t-1}

566 is the sample size of reproductive adults in year $t - 1$. The model is otherwise similar to the

567 point estimates model, except that we use year-specific intercepts (α_t) to incorporate the Poisson

568 offset (N_{t-1}), rather than a single model intercept as before. We fit the above-described models

569 again, with priors and settings as described in Supplementary Methods section S1.4.

570 For the stage-specific survival comparison shown in the climate figures, we derived annual stage-

571 specific survival probabilities as column sums of U (survive and transition to any living stage).

572 We then estimated temperature effects for each stage under two formulations on the logit scale:

573 an SSD-weighted quasibinomial GLM for point estimates ($\text{surv} \sim \text{tmp}$, with weights from the

574 stable stage distribution of each annual U matrix) and a binomial count model for uncertainty

575 propagation ($\text{cbind}(n_{\text{surv}}, n_{\text{fail}}) \sim \text{tmp}$). The two formulations also differ in how they weight the

576 data, handle dispersion, and use sample sizes, so comparing them does not single out the effect

577 of sampling uncertainty. Full implementation details are provided in Supplementary Methods

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 $b \sim \text{MVN}(\mu_b, \mathbf{K})$
 $K_{i,j} = \eta^2 \exp(-0.5 \rho^2 (j-i)^2)$ (14)

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sections S1.6 and S1.10. We also applied this approach to a multi-site dataset of three *Astragalus scaphoides* populations as a robustness check (Supplementary Methods section S1.10; Supplementary Results section S2.5).

Analysis 3: *Simulation study of climate effects on fecundity*

To evaluate the effect of sampling uncertainty where the true climate effect is known, we simulated annual recruitment time series from a realistic four-stage Lefkovitch MPM generated with `mpmsim::rand_lefko_mpm()` (archetype 2). We kept a baseline matrix with population growth rate close to equilibrium ($\lambda \approx 1$). We then imposed a climate effect on a single fecundity entry, $F_{1,4}$, over $T = 15$ annual time steps. In each simulated year, the climate covariate was drawn as $x_t \sim \text{Normal}(0,1)$, and true fecundity followed

$$\log(F_{1,4,t}) = \log(F_{1,4}) + \beta_{\text{true}}x_t + z_t, \quad (13)$$

where β_{true} was set to 0.1, 0.3, or 0.5, and $z_t \sim \text{Normal}(0, 0.2^2)$ represented additional year-to-year biological variation not explained by climate. For each year we then simulated observed seedling counts from finite samples of reproductive individuals, using stage sample sizes of 10, 20, 50, or 100 per year. We analysed 200 replicate time series for each effect-size and sample-size combination.

For each simulated dataset, we fitted three models to estimate the climate effect on recruitment: (i) a Gaussian regression with a log link on annual point estimates of fecundity, (ii) a Poisson count model using observed seedling counts and an offset for the number of reproductive individuals, and (iii) a Poisson uncertainty-aware model that additionally included a latent year-specific process term analogous to z_t . All three models target the same log-scale climate effect; the Gaussian model uses untransformed rates as its response and assumes constant residual variance. Its 95% intervals are the usual coefficient estimate $\pm t \times \text{standard error}$. We then compared the models in terms of bias and 95% interval coverage (the proportion of replicate

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Deleted: To test whether the analysis 2 conclusions generalise across populations, we applied the same paired modelling framework to three *Astragalus scaphoides* populations (Haynes Creek, McDevitt Creek, and Sheep Corral Gulch) (Crone & Lesica, 2004). For each site, we fit the same two recruitment-temperature formulations used in analysis 2: a point-estimate model and a sampling-uncertainty model, for both (i) monthly-lag coefficients in the Gaussian-process framework and (ii) a simplified spring-temperature model. We also fit the same paired stage-specific survival-temperature comparison used in analysis 2, deriving annual stage-specific survival from U -column sums and comparing an SSD-weighted quasibinomial point-estimate model with a binomial count model that propagates sampling uncertainty. This preserves comparable linear predictors across formulations while varying only how uncertainty enters the response model. Full implementation details are given in Supplementary Methods section S1.9.

633 datasets in which the 95% interval contained the true effect β_{true} ; a well-calibrated model
 634 achieves coverage close to the nominal 0.95). Both Poisson models allow for the sampling
 635 variation in counts; their comparison tests the effect of additionally modelling year-to-year
 636 process variation.
 637 Priors and fitting settings for the simulation model are given in Supplementary Methods section
 638 S1.4.

639 All analyses were run in R (4.6.1) using RStan (2.32.7), with data handling and plotting in
 640 tidyverse packages.

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641 Results

642 *Analysis 1: Comparative analysis of mortality trajectories*

643 We first look, across all derived traits, at how much more spread out the point estimates are than
 644 their posterior means across every population-level entry (Table 1). We then ask whether that
 645 extra spread is large enough to change the conclusions of a particular comparative analysis.
 646 For life expectancy and shape, across-entry variance of point estimates was 1.93 and 3.31 times
 647 the variance of posterior means, respectively (Table 1). These ratios describe how much more
 648 spread out the point estimates are, not how much of that spread comes from sampling noise
 649 rather than genuine biological differences among populations. Point estimates of life expectancy
 650 and shape often deviated from their respective sampling distributions (Fig. 3). For shape, the
 651 most extreme positive and negative point estimates tended to have sampling distributions shifted
 652 much closer to zero, whereas for mature life expectancy, many point estimates were skewed
 653 upward from the median of the sampling distribution. The mismatches between point estimates
 654 and sampling distributions usually coincided with a boundary estimate of stage-specific survival
 655 (i.e. a point estimate approaching either 0% or 100%). Supplementary analyses suggest that these

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667 boundary estimates can skew a point estimate of life expectancy or shape away from the majority
668 of its sampling distribution (see Supplementary Figure S1 and Supplementary Figure S2).

669 Despite the mismatch between point estimates and sampling distributions, accounting for

670 sampling uncertainty in our test of the relationship between life expectancy and shape ~~did not~~
671 ~~impact~~ the overall ~~conclusion~~ (Fig. 4; ~~Supplementary Table S3~~). ~~While~~ the posterior distribution
672 ~~of the slope parameter β~~ from the point-estimate model was centred slightly to the left of zero
673 ($P[\beta > 0] = 0.206$), the corresponding posterior from the sampling-uncertainty model was
674 shifted somewhat to the right ($P[\beta > 0] = 0.770$). ~~The posterior median of the slope therefore~~
675 ~~changed sign between the two models. This shift is not trivial, but because both posteriors span~~
676 ~~zero the overall conclusion, that there is no strong effect, is unchanged.~~

677 Counterintuitively, there was less posterior uncertainty in β in the sampling-uncertainty model

678 than in the point-estimate model. ~~The reason is largely~~ that a regression slope is harder to
679 estimate precisely when the ~~predictor is~~ very spread out, ~~since~~ many slopes ~~then~~ fit the data
680 ~~about equally well. Boundary-survival estimates push some shape point estimates to extreme~~
681 ~~values, artificially inflating that spread; replacing them with~~ their sampling distributions, which
682 cluster much closer to zero. ~~(Figure 4A), pulls the predictor back in and tightens the slope.~~
683 ~~Posterior width also depends on the model's assumptions and treatment of uncertainty.~~

684 *Analysis 2: The impact of weather on demographic traits*

685 Despite the relatively small sample sizes underlying some recruitment estimates (the three
686 smallest sample sizes of reproductive individuals were 8, 12, and 13, respectively), results from
687 the point-estimate model and the sampling-uncertainty model were virtually identical (Fig. 5).
688 Both models identified temperature deviations in monthly lags 7-9 (February-April, prior to the
689 start of the transition interval in July) as having a positive effect on recruitment. If we simplify
690 our model and consider only the relationship between recruitment and mean spring temperature
691 deviations (see Supplementary Methods section S1.6 and Supplementary Table S5), we again find

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713 that the point-estimate model is similar to the model incorporating sampling uncertainty (Fig. 5).

714 The estimate of the temperature coefficient β was 8.0% higher, and the 95% interval width for

715 that estimate was 10.9% higher, in the model incorporating sampling uncertainty compared with

716 the point-estimate model. A difference of this magnitude is unlikely to qualitatively change any

717 inferences drawn from a model. In the stage-specific survival panel (Fig. 5C), temperature

718 coefficients were estimable for stages 2 and 3; ~~stages 1 and 4 are labelled as not estimable~~

719 ~~because stage 1 has only four usable years and stage 4 has 100% survival in all usable years. The~~

720 ~~narrower intervals in the count model should be interpreted in light of the different likelihood~~

721 ~~and weighting assumptions, rather than attributed to uncertainty propagation alone. A~~

722 ~~supplementary multisite robustness analysis based on three *Astragalus scaphoides* populations~~

723 ~~reached the same qualitative conclusion: climate-effect inference was similar with and without~~

724 ~~explicit uncertainty propagation in that empirical setting (Supplementary Results section S2.5;~~

725 ~~Figure S5; Tables S7-S10).~~

726 *Analysis 3: Simulation study of climate effects on fecundity*

727 ~~The simulation study showed that accounting for year-to-year process variation improved the~~

728 ~~intervals' coverage without much increasing the average error (Fig. 6). Across all effect sizes and~~

729 ~~sample sizes, the uncertainty-aware Poisson model achieved coverage close to the nominal 95%~~

730 ~~(0.951), whereas the simpler Poisson point-estimate model's intervals fell well short (0.757). The~~

731 ~~two Poisson models were about equally accurate on average (RMSE 0.086 versus 0.087).~~

732 ~~Modelling this process variation therefore mainly improved how often the intervals contained~~

733 ~~the true effect, rather than the accuracy of the estimate itself.~~

734 ~~The Poisson point-estimate model's intervals missed the true effect more and more often as~~

735 ~~sample size increased. Averaged across true effect sizes, its intervals contained the true effect in~~

736 ~~0.897 of cases at sample size 10, falling to 0.593 at sample size 100, whereas the uncertainty-~~

737 ~~aware model stayed near 0.95 over the same range (0.955 to 0.952; Fig. 6). Larger samples shrink~~

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Deleted: Extending the same framework to three *Astragalus scaphoides* populations yielded the same overall pattern: point-estimate and sampling-uncertainty models were closely aligned within site, while differences among sites were more pronounced (Fig. 6). Across sites, we had 9-11 years of annual demographic records, with 8-10 years retained for model fitting after filtering (0-3 years excluded; Supplementary Table S7).[¶] For spring-temperature models, posterior medians differed only slightly between formulations (maximum absolute within-site difference: 0.026 on the linear predictor scale), and 95% interval widths changed by only -5.3% to 4.7% across sites (Supplementary Table S8). Monthly-lag coefficient profiles were likewise highly concordant between formulations, with within-site correlations of posterior medians ranging from 0.992 to 0.997 and a maximum lag-specific median difference of 0.042 (Fig. 6). As in analysis 2, these results indicate limited sensitivity of climate-effect inference to transition-rate sampling uncertainty in this setting.[¶]

the sampling noise in each year's estimate, so the point-estimate model treats its estimate as increasingly precise and its intervals narrow. But the simulated truth also varies from year to year for reasons unrelated to climate, and this variation does not shrink with sample size. The point-estimate model ignores it, so its intervals become too narrow to capture the true effect, increasingly so as samples grow, whereas the uncertainty-aware model includes a term for it and keeps its intervals about the right width. This year-to-year variation is a different source of uncertainty from the sampling noise in the transition rates that we examined in Analyses 1 and 2. On average, all three models estimated the climate effect with little systematic error (Fig. 6). For the Gaussian log-link model, the average estimate of the strongest effect ($\beta_{\text{true}} = 0.5$) was off by only -0.005, but its 95% intervals contained the true effect only 0.915 of the time, below the intended 0.95. The uncertainty-aware model therefore gives both an accurate estimate and intervals that contain the true effect close to 0.95 of the time.

Discussion

When does transition-rate sampling uncertainty materially affect inference from published MPMs?

Sampling uncertainty in the transition rates underlying matrix population models can, in principle, propagate into substantial uncertainty in the demographic traits derived from those models. However, our results show that the impact of sampling uncertainty on inference from published MPMs is strongly trait-dependent. In our analyses, accounting for transition-rate sampling uncertainty moderately affected inference in a comparative analysis of mortality trajectories, but had little impact on a within-species analysis of weather effects on recruitment.

The simulation analysis clarifies that this apparent robustness does not imply that point-estimate count models are generally well calibrated: when unmodelled year-to-year biological variation is present, they can remain nearly unbiased yet still become overconfident. It is important to note that the within-species result pertains specifically to an aggregate reproductive transition

796 (seedlings per reproductive adult) rather than to stage-specific survival transitions, which are
 797 more commonly affected by boundary estimates. The contrast between analyses 1 and 2,
 798 together with the simulation result, therefore reflects both the type of trait analysed and the type
 799 of transition underlying it.

800 We use “materially affects inference” to mean a difference large enough to change a qualitative
 801 conclusion, for instance, by shifting the direction of an estimated effect, moving a credible
 802 interval from largely overlapping to largely excluding zero, or substantially altering the rank order
 803 of populations. This is a judgment call rather than a formal threshold, and readers with different
 804 tolerances for uncertainty may draw the lines differently. This contrast illustrates that ignoring
 805 sampling uncertainty does not universally compromise inference from published projection
 806 matrices, but that its consequences depend on the traits being analysed and the structure of the
 807 underlying models.

808 Point estimates were far more dispersed relative to their posterior means for mortality-trajectory
 809 traits, particularly shape, than for integrated metrics, such as population growth rate (λ) and
 810 generation time (Table 1). This greater dispersion, together with the displacements shown in
 811 Figure 3, indicates that mortality-trajectory traits are more sensitive to sampling uncertainty, than
 812 the directly observed recruitment response in the climate analysis.

813 *Why are some MPM-derived traits particularly sensitive to sampling uncertainty, and what are the practical*
 814 *implications?*

815 The sensitivity of mortality-trajectory traits has a clear mechanistic basis. Life expectancy and
 816 shape are inferred via age-from-stage methods, which project a cohort forward through repeated
 817 matrix multiplication, so errors in late-stage survival are not simply carried forward but
 818 compounded across projection steps, amplifying small transition-rate uncertainties into large
 819 uncertainties in long-term survivorship. Integrated metrics such as population growth rate are

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Deleted: , and individual growth rate, sampling uncertainty accounted for only a small fraction of

Deleted: total variation observed among populations. By contrast, traits related to mortality trajectories, particularly the shape of the

Deleted: distribution, were far

Deleted: : nearly half of the total variation in shape was attributable...

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844 buffered against this, because they average over many transitions where errors tend to partially
 845 cancel, although they are not wholly immune to sampling uncertainty.
 846 Several structural features of published matrices may exacerbate this. In our Analysis 1 life
 847 expectancy-shape set (68 population-level matrix entries from 25 species), 47% (32/68) contain
 848 only a single reproductive stage class, 69% (47/68) reach their quasi-stable distribution in less
 849 than one life expectancy, and 46% (31/68) contain at least one stage-specific survival estimate of
 850 100% after rescaling any U columns whose sums exceeded 1 (allowing for floating-point
 851 rounding). These are not rare edge cases; they are common features in our dataset that may
 852 contribute to displacement of derived trait estimates from the centre of their sampling
 853 distributions.

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854 Consistent with this explanation, within the Analysis 1 dataset displacement magnitude increases
 855 with both the number of boundary survival stages and the proportion of sampled individuals
 856 occupying those stages (Supplementary Methods section S1.12; Table S13; Figure S3).

857 *Practical implications for users of compiled demographic databases*

858 These structural constraints have practical implications. Highly aggregated demographic metrics,
 859 such as population growth rate and generation time, were among the least sensitive in our
 860 dataset, so analyses focused on them will often be robust to sampling uncertainty. Analyses of
 861 ageing, longevity, or mortality trajectories were more sensitive and warrant particular caution,
 862 Generalisability is also conditioned by taxonomic representation. Relative to the full current
 863 COMPADRE plant database, our final 31-study subset is enriched for herbaceous perennials
 864 (64.5% of studies in our subset versus 49.1% in COMPADRE overall) and succulents (16.1%
 865 versus 7.8%), while trees (6.5% versus 15.4%) and shrubs (3.2% versus 9.0%) are under-
 866 represented; annuals and several less common plant life forms are absent entirely. This skew
 867 mirrors a documented bias in plant demographic data more broadly, towards short-term studies

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888 of temperate-region herbaceous perennials (Römer et al. 2024). Our conclusions should
889 therefore be interpreted most directly for perennial herb and succulent systems, and more
890 cautiously for woody or otherwise under-represented life forms.

891 Within the analysed subset, the warning signs most relevant to uncertainty propagation were also
892 unevenly distributed. Boundary survival estimates and very small stage counts were concentrated
893 mainly in herbaceous perennial systems and in larger matrices: among 6+-stage population-level
894 entries, 61.1% contained at least one boundary survival estimate and 44.4% had at least one stage
895 with $N < 20$, compared with 36.0% and 26.0%, respectively, for 3-5-stage entries (Table S12).
896 This pattern is consistent with the practical intuition that more weakly observed stage classes
897 create more opportunities for boundary estimates, although woody life forms remain too sparse
898 in our final subset for a strong life-form-specific comparison (Supplementary Methods section
899 S1.11).

900 As provisional screening heuristics, analyses warrant particular caution when: any stage has
901 survival >0.95 or <0.05 , or is estimated from fewer than 20 individuals; the matrix has only a
902 single reproductive stage class; or the time to quasi-stable distribution is less than one life
903 expectancy. Conversely, analyses ~~may be less sensitive~~ when survival estimates are well away
904 from boundaries, key transitions are estimated from >50 individuals, and the focal metric is
905 highly integrated (e.g. population growth rate, generation time). The absence of a detectable
906 uncertainty effect (as in our climate-recruitment analysis) should not be taken as general evidence
907 that propagation is unnecessary; robustness is trait- and context-specific.

908 Our analyses used count-based, stage-structured (Lefkovich) matrices, and the consequences of
909 sampling uncertainty may differ for other widely used demographic model types. In age-based
910 (Leslie) models, vital rates are indexed by age rather than stage, so the age-from-stage
911 reconstruction that amplifies late-stage survival errors in our Analysis 1 does not arise in the
912 same way, but uncertainty can still accumulate when survival is projected across ages. Analyses

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built directly from individual-level data, such as integral projection models or capture-recapture survival models, typically estimate vital rates by sharing information across ages or sizes; such partial pooling tends to reduce the variance of weakly informed rates but can induce correlations among them, altering both the magnitude and the structure of the uncertainty that propagates to derived quantities. In those settings, propagating uncertainty from the joint posterior of the fitted vital-rate model, rather than treating rate estimates as independent, would be the natural extension of our approach.

Sampling uncertainty versus demographic stochasticity

It is important to emphasise that our analysis of sampling uncertainty is distinct from demographic stochasticity, a distinction that is sometimes blurred in the literature (Kendall 1998). In the current study, we assume that the parameter of interest is the population mean of a demographic trait and focus exclusively on uncertainty arising from finite sampling of that mean. In addition, individuals within a population are likely to exhibit genuine variation around the population mean in demographic rates, giving rise to demographic stochasticity. Both sources of uncertainty can have important implications for applications such as population viability analyses, where failing to account for either demographic stochasticity or sampling uncertainty can lead to biased estimates of extinction risk (Kendall and Fox 2002, McGowan et al. 2011). Distinguishing between these sources of uncertainty is therefore essential for appropriate interpretation.

Data limitations and reporting considerations

Incorporating transition-rate sampling uncertainty into analyses of published MPMs remains challenging, primarily because the data required to reconstruct sampling distributions are rarely reported. In our own screening, most otherwise-eligible studies had to be excluded for exactly this reason. Improved reporting of transition-specific sample sizes or standard errors, particularly on the scale of the linear predictor, would greatly enhance future researchers' ability to assess

Deleted: Of the 86 source publications we initially targeted, only a small subset provided sufficient information to model uncertainty in survival or reproductive transitions.

942 robustness and propagate uncertainty where necessary (see Supplementary Methods section S1.7
943 for detailed recommendations). At the same time, the absence of a single, universally “correct”
944 approach to modelling transition rates should not deter progress. Even partial information can
945 be sufficient to identify cases where inference is likely to be robust or fragile, and recognising the
946 limits of inference is often as important as refining point estimates. Viewed in this light,
947 accounting for sampling uncertainty is not simply a technical exercise but a means of
948 understanding when demographic inference from compiled databases can be trusted and when it
949 should be treated with caution.

950 *Conclusions*

951 Our analyses show that the consequences of transition-rate sampling uncertainty in published
952 matrix population models are highly trait-dependent. For integrated metrics such as population
953 growth rate and generation time, point estimates were only modestly more dispersed than their
954 posterior means, whereas for mortality-trajectory traits, including life expectancy and especially
955 the shape of the mortality distribution, they were much more dispersed and more sensitive to
956 sampling uncertainty. In these cases, point estimates can be displaced from the centres of their
957 posterior distributions by boundary estimates of stage-specific survival.

958 These results indicate that accounting for sampling uncertainty will not universally alter
959 conclusions drawn from analyses of published MPMs, but that its importance depends critically
960 on the demographic traits being analysed and on features of model structure and data availability.
961 Although our study examined only a subset of MPM-derived parameters and a limited fraction
962 of the matrices available in COMPADRE, it highlights specific conditions under which inference
963 is likely to be fragile and provides practical guidance for assessing robustness in comparative and
964 time-series analyses.

965 More broadly, the growing availability of large, compiled demographic databases presents both
966 new opportunities and new challenges. The data underlying these databases were collected for

- Deleted: many commonly used
- Deleted: demographic
- Deleted: ,
- Deleted: sampling uncertainty accounted for
- Deleted: a small fraction of total variation, suggesting that inference from published MPMs is often robust for these traits. By contrast, parameters related to
- Deleted: trajectories
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- Deleted: systematically biased

982 diverse purposes, using heterogeneous sampling designs and statistical approaches, resulting in
983 variation not only in sampling uncertainty but also in underlying assumptions and context.
984 Recognising and accounting for these limitations through improved reporting in primary studies
985 and careful, trait-specific interpretation in secondary analyses will be essential to ensure that
986 comparative demography based on published projection matrices yields reliable and biologically
987 meaningful inferences.

988 **Acknowledgements**

989 [removed for anonymous review]

990 **Author contributions**

991 [removed for anonymous review]

992 **Conflict of interest**

993 The authors declare no conflict of interest.

994 **Data availability**

995 Data and code required to reproduce our analyses are provided for review in a zip archive. They
996 will be archived in a publicly accessible repository upon manuscript acceptance.

997

998 **Supplementary Materials**

999 The supplementary materials include the following items:

- 1000
- **Supplementary Methods S1.1:** Sampling-distribution workflow used for each MPM.

- 1001 • **Supplementary Methods S1.2:** Age-from-stage derivation, truncation, and shape
- 1002 metric.
- 1003 • **Supplementary Methods S1.3:** Pooled and mean MPM handling.
- 1004 • **Supplementary Methods S1.4:** Bayesian model priors and fitting settings.
- 1005 • **Supplementary Methods S1.5:** [Comparison of point-estimate and posterior-mean](#)
- 1006 [variance](#).
- 1007 • **Supplementary Methods S1.6:** Simplified spring-temperature model ([Analysis 2](#)).
- 1008 • **Supplementary Methods S1.7:** Reporting recommendations for future MPM studies.
- 1009 • **Supplementary Methods S1.8:** [Included versus excluded studies in the current target-](#)
- 1010 [study screen](#).
- 1011 • **Supplementary Methods S1.9:** [Species identities in Analysis 1](#).
- 1012 • **Supplementary Methods S1.10:** Multisite robustness extension [of Analysis 2](#) (*Astragalus*
- 1013 *scaphoides*).
- 1014 • **Supplementary Methods S1.11:** Boundary survival estimates and small stage-specific
- 1015 [sample sizes in Analysis 1](#).
- 1016 • **Supplementary Methods S1.12:** Displacement magnitude versus boundary survival
- 1017 [structure](#).
- 1018 • **Supplementary Methods S1.13:** Sensitivity to the quasi-stable-distribution truncation
- 1019 [threshold](#).
- 1020 • **Supplementary Results S2.1:** Sensitivity to point-estimate weighting choice (no weights
- 1021 vs SSD weights).
- 1022 • **Supplementary Results S2.2:** [Where do boundary estimates and small stage counts](#)
- 1023 [occur?](#)

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- 1030 • [Supplementary Results S2.3: Does displacement increase with boundary survival](#)
- 1031 [structure?](#)
- 1032 • [Supplementary Results S2.4: Sensitivity of shape estimates to the truncation threshold.](#)
- 1033 • [Supplementary Results S2.5: Multisite robustness analysis in *Astragalus scaphoides*.](#)
- 1034 • **Table S1:** Single-MPM derived-parameter summary for the focal *Agrimonia eupatoria*
- 1035 example.
- 1036 • **Table S2:** Mean-MPM derived-parameter summary for the focal *Agrimonia eupatoria*
- 1037 example (population B).
- 1038 • **Table S3:** Analysis 1 posterior summary for the slope parameter (β) in the life
- 1039 expectancy-shape model.
- 1040 • **Table S4:** Analysis 2 Gaussian-process regression coefficients (monthly lags) for
- 1041 recruitment responses to temperature anomalies.
- 1042 • **Table S5:** Analysis 2 posterior summary for the spring-temperature coefficient.
- 1043 • **Table S6:** Primary studies represented in the analysis subset, as identified in
- 1044 COMPADRE.
- 1045 • **Table S7:** [Multisite robustness-analysis](#) site coverage and spring-temperature effect
- 1046 summaries for *Astragalus scaphoides*.
- 1047 • **Table S8:** [Multisite robustness-analysis](#) posterior summary for the spring-temperature
- 1048 coefficient by site and model formulation.
- 1049 • **Table S9:** Analysis 2 stage-specific survival-temperature coefficients under no weights
- 1050 and SSD weights.
- 1051 • **Table S10:** [Multisite robustness-analysis](#) stage-specific survival-temperature coefficients
- 1052 under no weights and SSD weights.

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- [Table S11: Comparison of screened target studies that were included versus excluded under the current reproducible COMPADRE-based screen.](#)
- [Table S12: Distribution of boundary survival estimates and small stage-specific sample sizes across Analysis 1 population-level MPM entries.](#)
- [Table S13: Linear models relating displacement magnitude in Analysis 1 to boundary-survival predictors.](#)
- [Table S14: Sensitivity of Analysis 1 shape estimates to the quasi-stable-distribution truncation threshold.](#)
- **Figure S1:** Boundary-estimate diagnostic.
- **Figure S2:** Illustration of survivorship trajectories under boundary survival transitions.
- [Figure S3: Displacement magnitude in Analysis 1 versus boundary-survival structure.](#)
- [Figure S4: Sensitivity of Analysis 1 shape estimates to the quasi-stable-distribution truncation threshold.](#)
- [Figure S5: Multisite robustness analysis for *Astragalus scaphoides*.](#)

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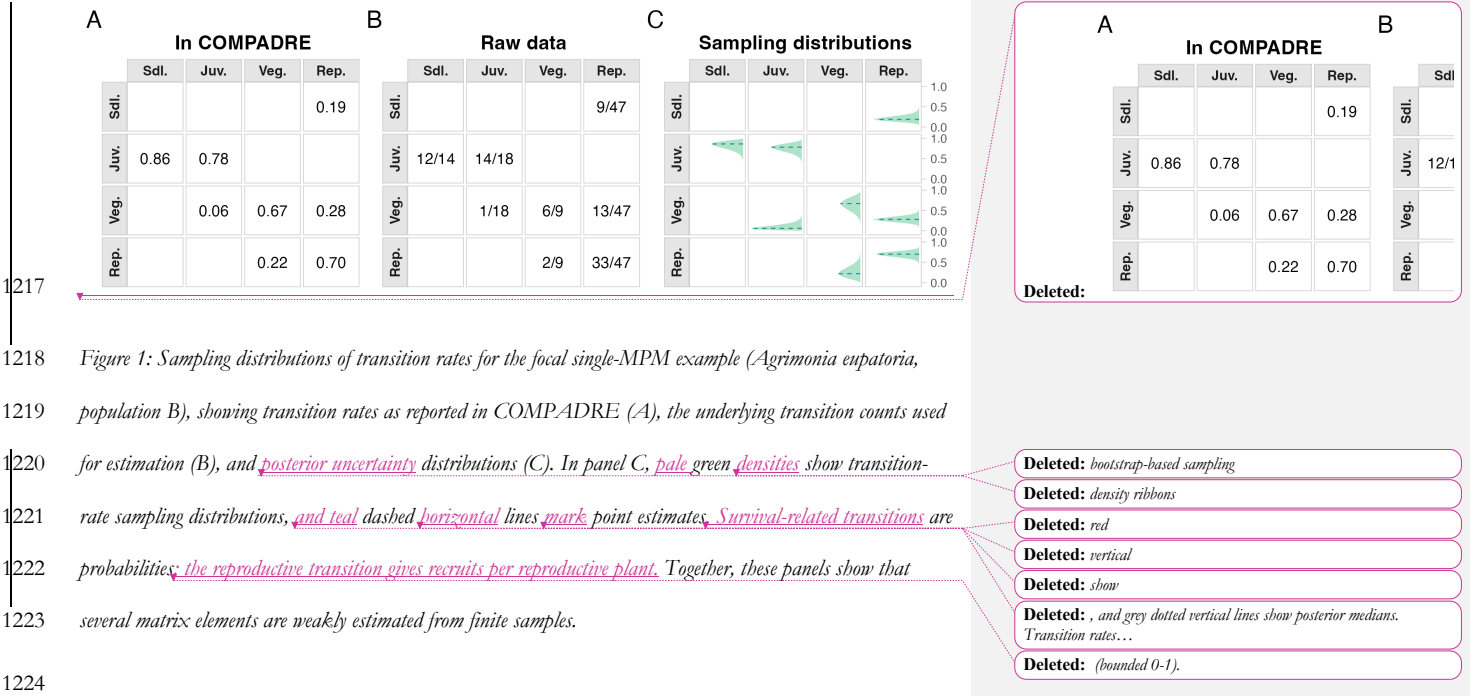
1199 **Tables**

1200 Table 1: *Across-entry* variance ratios for key demographic parameters in *Analysis 1*, with 95% intervals from
1201 *paired bootstrap resampling of population-level entries*. Each ratio is the variance of matrix point estimates divided
1202 *by the variance of reconstructed posterior means across the same entries*; values above 1 indicate that point
1203 *estimates are more spread out than posterior means, not how much of that spread is due to sampling noise rather*
1204 *than genuine biological differences*. The ratio is highest for shape, intermediate for life expectancy, and lowest for log
1205 population growth rate.

Parameter	Variance ratio	95% interval
Shape	3.31	[2.26, 5.06]
Life expectancy	1.93	[1.48, 2.55]
Log population growth rate	1.01	[0.99, 1.15]
Damping ratio	1.12	[1.01, 1.29]
Generation time	1.63	[1.20, 2.22]
Probability of reaching maturity	1.62	[1.08, 2.59]

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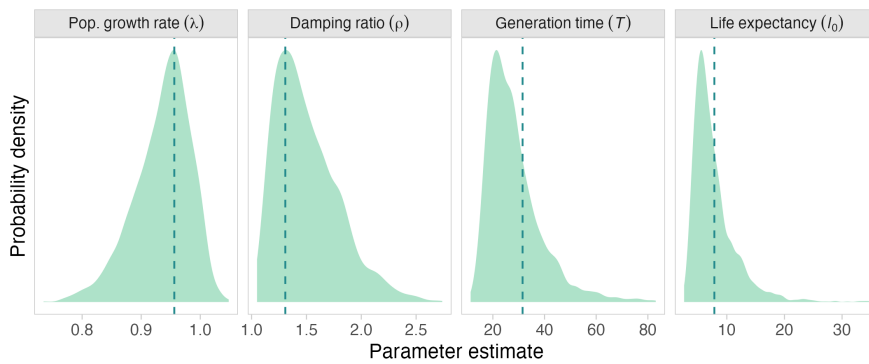
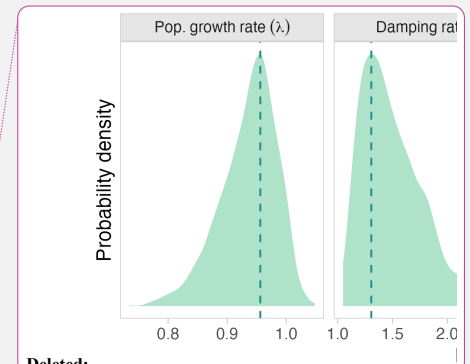


Figure 2: Sampling distributions of demographic quantities derived from the focal single MPM, including population growth rate (λ), damping ratio (ρ), generation time (T), and life expectancy (l_0). Pale green densities show derived-quantity sampling distributions, and teal dashed vertical lines mark point estimates. Quantities are shown on their natural scales (T and l_0 are in years). Uncertainty in transition rates propagates to substantial uncertainty in these derived parameters, with the largest spread in quantities that depend on long-term survivorship and reproduction.



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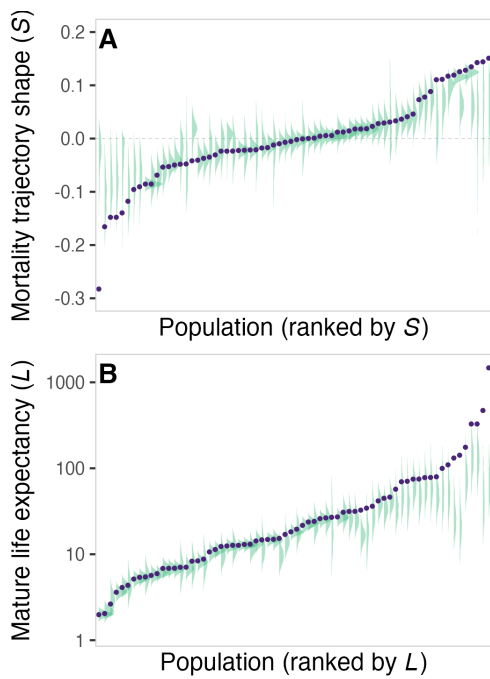
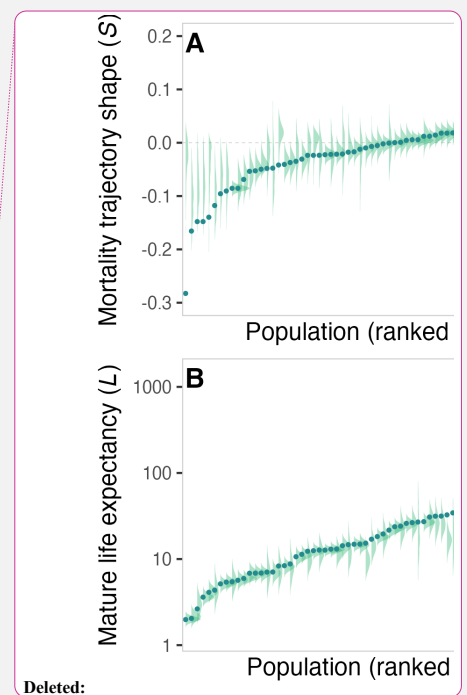


Figure 3: Analysis 1 comparison of point estimates versus sampling-distribution summaries for population-level life expectancy and shape. Panel A shows mortality-trajectory shape, and panel B shows mature life expectancy (years). Purple points show estimates derived directly from the matrix point estimates, whereas green distributions show uncertainty-aware summaries from transition-rate sampling distributions. The figure shows systematic shifts and shrinkage in extreme values when sampling uncertainty is propagated.



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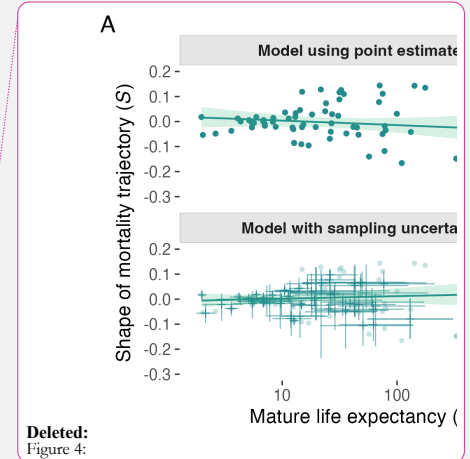
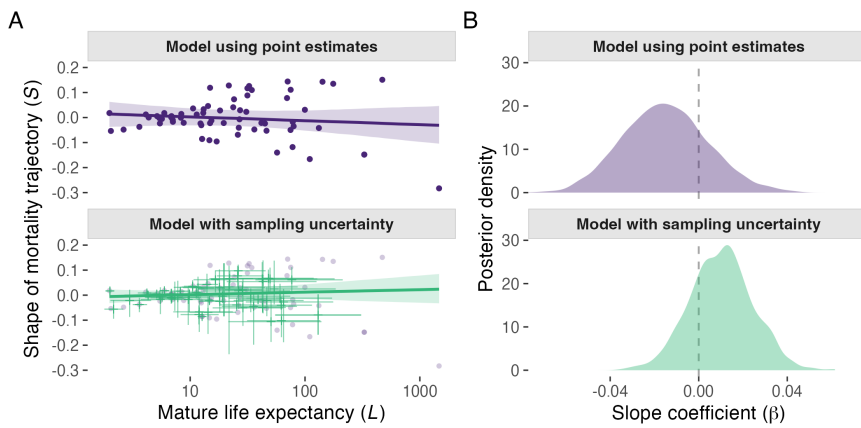
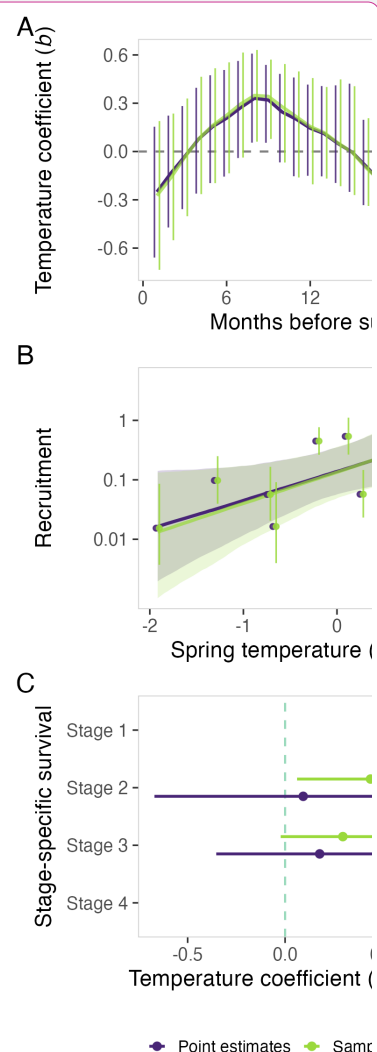
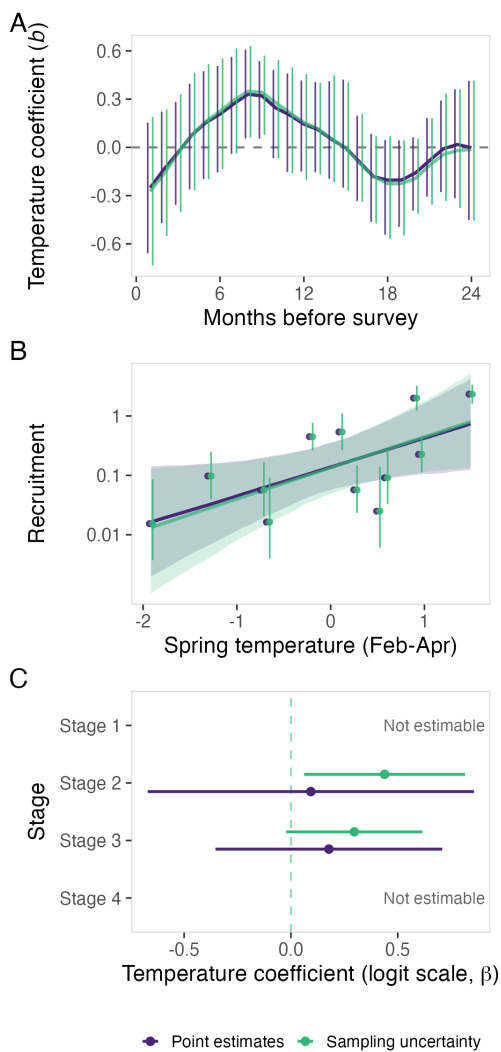


Figure 4: Accounting for sampling uncertainty shifts the estimated life expectancy-shape relationship and narrows its posterior interval, although both models retain substantial uncertainty about the direction of the relationship. Analysis 1 relationship between life expectancy and shape under models with and without sampling uncertainty propagation. In panel A, purple points, line, and ribbon show the model using point estimates. Green crosses, line, and ribbon show the uncertainty-aware model; crosses are centred on sampling-distribution medians with horizontal and vertical bars showing 95% intervals. The displacement between faint reference points and crosses indicates that propagated sampling distributions can shift central tendency away from point estimates, consistent with the offsets between dashed lines and density peaks in Figure 2. Axes are mature life expectancy (years, log scale) and mortality-trajectory shape. Panel B shows posterior densities for the slope parameter β (unit change in shape per unit change in log life expectancy), with purple for the point-estimate model and green for the uncertainty-aware model.

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Figure 5: Analysis 2 climate effects on recruitment and stage-specific survival in *Silene spaldingii*. Panels A-B show monthly-lag and spring-temperature recruitment models for point-estimate (purple) and uncertainty-aware (green) formulations, with similar posterior coefficient estimates and interval widths. In panel A, y-values are temperature coefficients b for monthly temperature deviations (months before survey on the x-axis). In panel B,

1288 recruitment is seedlings per reproductive plant and spring temperature is February-April temperature deviation.

1289 Panel C shows stage-specific survival-temperature coefficients on the logit scale ⁽³⁾ where estimable (stages 2-3),

1290 again with purple for the point-estimate model and green for the uncertainty-aware model. Panel C intervals are

1291 95% Wald confidence intervals; differences reflect weighting and dispersion assumptions as well as the use of

1292 sample sizes. Stages 1 and 4 are labelled as not estimable (only four usable years and 100% survival in all usable

1293 years, respectively).

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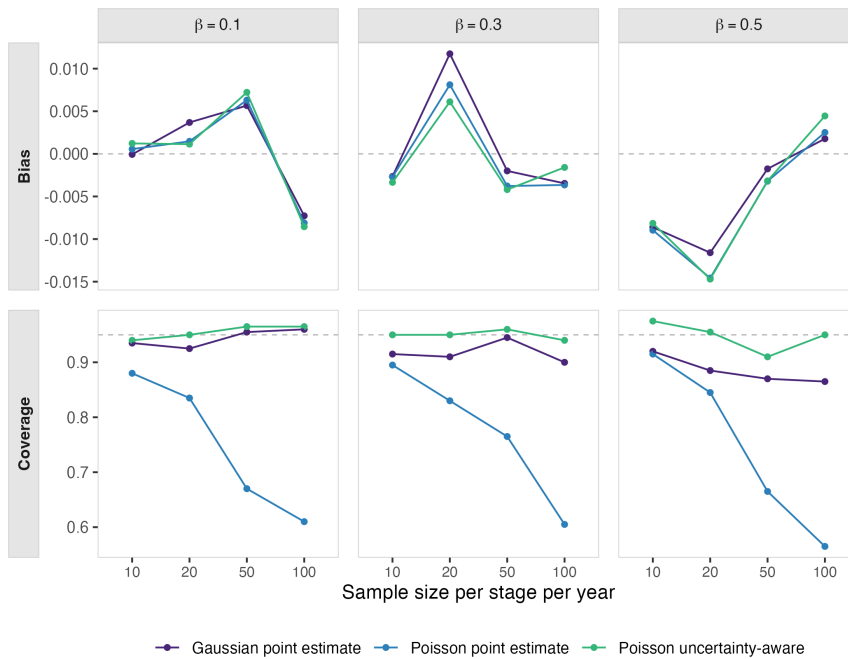
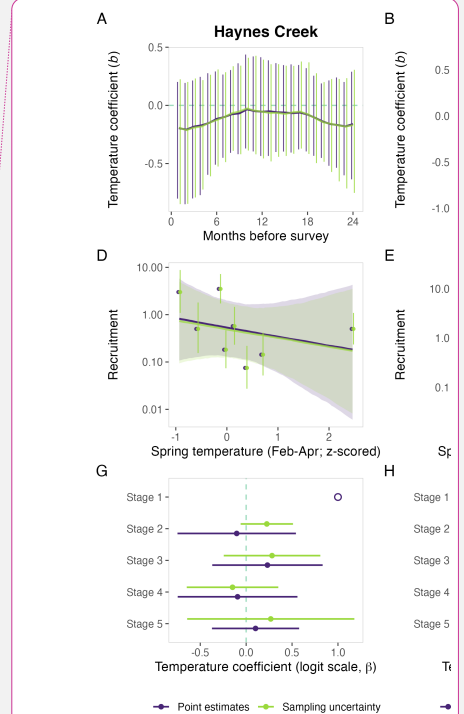


Figure 6: Simulation analysis of climate effects on fecundity, showing that modelling year-to-year process variation improves interval performance while keeping bias low. The upper row shows bias, and the lower row shows 95% interval coverage, across four sample sizes and three true effect sizes ($\beta = 0.1, 0.3, \text{ and } 0.5$). The Gaussian log-link point-estimate model (purple) has little average bias but its intervals are too narrow, especially at stronger climate effects. The Poisson point-estimate model (blue) is nearly unbiased but its intervals miss the true effect too often, especially at larger sample sizes. The uncertainty-aware Poisson model (green) keeps coverage close to the expected 95% level across the full design.



Deleted: Figure 6: Analysis 3 multisite climate-demography comparison for *Astragalus scaphoides*. Panel A (Haynes Creek), panel B (McDevitt Creek), and panel C (Sheep Corral Gulch) each show monthly-lag temperature coefficients, recruitment-temperature fits, and stage-specific survival-temperature coefficients for point-estimate (purple) and uncertainty-aware (green) formulations. Recruitment is seedlings per reproductive plant, and stage-specific survival coefficients are on the logit scale (β). Across sites, the two formulations remain closely aligned, showing that among-site ecological differences are larger than differences induced by uncertainty propagation.